

note

Dongkun Zhang

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1 Markov Decision Process

(Definition) Return *Something*

$$g_{t_0} = \sum_{t=t_0}^{\infty} \gamma^{t-t_0} r(s_t, a_t)$$

(Definition) Index and Goal *the agent's goal is to obtain a policy which maximises the cumulative discounted reward from $t = 0$:*

$$J(\pi, \gamma, p_0, p) = \mathbb{E}[g_0; \pi, \gamma, p_0, p]$$
$$\max_{\pi} J(\pi, \gamma, p_0, p)$$

(Proposition) *Define $\Pr(s \rightarrow s', k, \pi)$ as the probability of transitioning from state s to state s' in k steps under policy π , then*

$$\begin{aligned} J(\pi, \gamma, p_0, p) &= J(\pi, \rho^{\pi}) \\ &= \mathbb{E}_{s \sim \rho^{\pi}(\cdot), a \sim \pi(\cdot|s)}[r(s, a)] \\ &= \sum_s \rho^{\pi}(s) \sum_a \pi(a|s) r(s, a) \end{aligned}$$

where $\rho^{\pi}(s) = \sum_{s_0} p_0(s_0) \sum_{t=0}^{\infty} \gamma^t \Pr(s_0 \rightarrow s, t, \pi)$ is the (improper) discounted state distribution.

Proof.

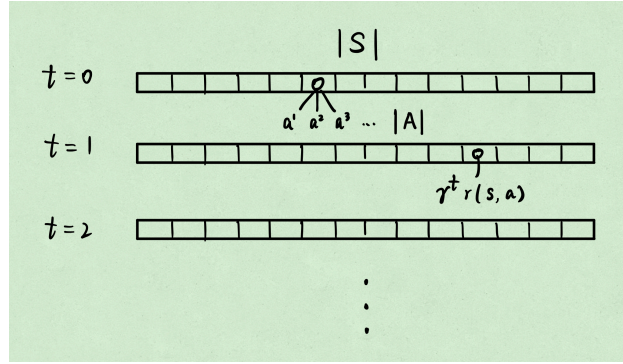


Figure 1: A MDP table.

Consider a table $\mathcal{T} \in \mathbb{R}_+ \times \mathbb{R}^{|S| \times |A|}$ as shown in Fig. 1. Therefore,

$$\begin{aligned}
 J(\pi, \gamma, p_0, p) &= \mathbb{E}[g_0; \pi, \gamma, p_0, p] \\
 &= \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t); \pi, p_0, p\right] \\
 &= \sum_{s_0} p_0(s_0) \sum_{t=0}^{\infty} \gamma^t \sum_s \sum_a \pi(a|s) \Pr(s_0 \rightarrow s, t, \pi) r(s, a) \\
 &= \sum_s \rho^\pi(s) \sum_a \pi(a|s) r(s, a) \\
 &= \mathbb{E}_{s \sim \rho^\pi(\cdot), a \sim \pi(\cdot|s)}[r(s, a)]
 \end{aligned}$$

End proof. ■

$$\begin{aligned}
 V^\pi(s) &= \mathbb{E}[g_t | s_t = s; \pi, \gamma, p] \\
 Q^\pi(s, a) &= \mathbb{E}[g_t | s_t = s, a_t = a; \pi, \gamma, p]
 \end{aligned}$$

2 Value Function

$$\begin{aligned}
 V^\pi(s) &= \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^\pi(s, a)] \\
 Q^\pi(s, a) &= r(s, a) + \mathbb{E}_{s' \sim p(\cdot|s, a)}[V^\pi(s')]
 \end{aligned}$$

$$\begin{aligned}
 V(s) &= V^*(s) = \max_{\pi} V^\pi(s) \\
 Q(s, a) &= Q^*(s, a) = \max_{\pi} Q^\pi(s, a)
 \end{aligned}$$

$$V(s) = \max_a Q^*(s, a)$$

Value Function *Something*

$$\begin{aligned} Q(s, a) &= r(s, a) + \mathbb{E}_{s' \sim p(\cdot|s, a)}[V(s')] \\ &= r(s, a) + \mathbb{E}_{s' \sim p(\cdot|s, a)}[\max_{a'} Q(s', a')] \end{aligned}$$

Proof.

Before

$$\begin{aligned} Q(s, a) &= \max_{\pi} Q^{\pi}(s, a) \\ &= r(s, a) + \max_{\pi} \mathbb{E}_{s' \sim p(\cdot|s, a)}[V^{\pi}(s')] \end{aligned}$$

After. ■

(Definition) Advantage Function *Something*

$$\begin{aligned} A^{\pi}(s, a) &= Q^{\pi}(s, a) - V^{\pi}(s) \\ \mathbb{E}_{a \sim \pi(\cdot|s)}[A^{\pi}(s, a)] &= 0 \end{aligned}$$

Optimal Advantage Function *Something*

$$\begin{aligned} A(s, a) &= Q(s, a) - V(s) \\ A(s, a^*) &= 0, \quad a^* = \arg \max_a Q(s, a) \end{aligned}$$

Proof.

Before

$$\begin{aligned} V^{\pi}(s) &= \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^{\pi}(s, a)] \\ &= \mathbb{E}_{a \sim \pi(\cdot|s)}[A^{\pi}(s, a) + V^{\pi}(s)] \\ &= \mathbb{E}_{a \sim \pi(\cdot|s)}[A^{\pi}(s, a)] + V^{\pi}(s) \end{aligned}$$

After. ■